

A CROSS-ENTROPY NOISE PROPAGATION OPERATOR AND THE EFFECTIVE RANK AT THE NEURAL COLLAPSE FIXED POINT

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ABSTRACT. We extend the noise-propagation framework of [1]—originally formulated for the squared loss in regression—to multi-class classification under the cross-entropy loss. The technical step is to derive a Fisher-corrected noise propagation operator $\mathbf{M}_{\text{CE}}$ directly from the cross-entropy Hessian $\nabla_{\theta}^2 \ell_{\text{CE}} = \mathbf{J}^\top \mathbf{H} \mathbf{J}$, where $\mathbf{H}(\mathbf{x}) = \text{diag}(\mathbf{p}(\mathbf{x})) - \mathbf{p}(\mathbf{x})\mathbf{p}(\mathbf{x})^\top$ is the output-space Fisher matrix at the softmax probabilities $\mathbf{p}(\mathbf{x})$. Our main result (Theorem 4.1) is that $\text{tr}(\mathbf{M}_{\text{CE}}) < \infty$ if and only if the underlying power-law exponents satisfy $\beta > \alpha + 1/2$, identical to the trichotomy for MSE. We then study the limit of $\mathbf{M}_{\text{CE}}$ along a sequence of trained predictors approaching the Neural Collapse fixed point of Papayan–Han–Donoho. Although $\mathbf{H}$ becomes singular in the strict one-hot limit, the limiting operator that controls the test-risk effective rank is well-defined on the $(K - 1)$ -dimensional zero-sum subspace fixed by the simplex equiangular tight frame, and Theorem 5.5 computes the effective rank of the per-sample block (equivalently, of the output-space averaged operator $\bar{\mathbf{M}}_{\text{CE}}$ on the zero-sum subspace $\mathcal{Z}_K$): it equals exactly $K - 1$, the number of classes minus one. The argument is purely linear-algebraic given the simplex ETF geometry, and yields a dimension-free generalization scaling for K -class classification at the terminal phase of training.

1. INTRODUCTION

The unified bound of [1] on the test risk of overparameterized regression involves a noise-propagation operator $\mathbf{M} = \Sigma^{-1} \Gamma \Sigma^{-1}$ built from the SVD of the training Jacobian $\mathbf{J}$ and the test–train cross-correlation Γ of gradient features. The trichotomy $\text{tr}(\mathbf{M}) < \infty \iff \beta > \alpha + 1/2$,

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under power-law decay $\sigma_j \asymp j^{-\alpha}$ and $\varrho_j := \sqrt{\Gamma_{jj}} \asymp j^{-\beta}$, characterizes benign, tempered, and catastrophic regimes for squared loss; the power-law model is standard in the random-feature analysis of [8], with bias–variance refinements in the companion [2].

The same generalization story should apply to classification, but the squared-loss derivation does not directly transfer: cross-entropy introduces a nonlinear softmax link, and its parameter Hessian is $\mathbf{J}^\top \mathbf{H} \mathbf{J}$ rather than $\mathbf{J}^\top \mathbf{J}$. The natural question is whether a Fisher-corrected operator preserves the trichotomy and admits an exact effective-rank calculation at the Neural Collapse (NC) fixed point of [10].

This paper answers both questions affirmatively. We define $\mathbf{M}_{\text{CE}}$ as the natural generalization of $\mathbf{M}$ obtained by replacing $\mathbf{J}^\top \mathbf{J}$ with the cross-entropy Hessian (see Definition 3.1 and the derivation that precedes it). We then show:

- (i) The CE trichotomy $\text{tr}(\mathbf{M}_{\text{CE}}) < \infty \iff \beta > \alpha + 1/2$ holds whenever the predicted probabilities are bounded away from the one-hot vertices uniformly along the relevant directions (Theorem 4.1).
- (ii) Approaching the NC simplex equiangular tight frame fixed point, each per-sample block of $\mathbf{M}_{\text{CE}}$ on the zero-sum subspace has effective rank exactly $K - 1$ (Theorem 5.5), where K is the number of classes; the same holds for the averaged operator $\overline{\mathbf{M}}_{\text{CE}}$.

The first result extends the trichotomy across loss functions; the second states that, at the terminal phase of training, the generalization-controlling rank is set by the number of classes minus one, regardless of the input dimension. We work conditional on the presence of NC and characterize what this implies for the effective rank, rather than proving NC.

Relation to prior work. The Neural Collapse phenomenon was identified empirically in [10] and characterized mathematically in the unconstrained-features setting under cross-entropy by [7, 9] and under squared loss / central path by [6]. The class-imbalanced extension of unconstrained features is [5]. The cross-entropy Fisher matrix $\mathbf{H} = \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$ and its degeneracy at one-hot vertices is classical (see [4]); we use only the finite-rank facts about $\mathbf{H}$ and the simplex ETF geometry.

Organization. Section 2 fixes notation and recalls the classification setting. Section 3 derives $\mathbf{M}_{\text{CE}}$ from the CE Hessian. Section 4 proves the CE trichotomy. Section 5 derives the effective-rank formula at the NC fixed point. Section 6 discusses limitations.

2. SETTING AND NOTATION

Let $K \geq 2$ be the number of classes. The predictor has K -dimensional output $f(\mathbf{x}; \theta) = (f_1(\mathbf{x}; \theta), \dots, f_K(\mathbf{x}; \theta)) \in \mathbb{R}^K$, softmax probabilities $p_k(\mathbf{x}; \theta) = e^{f_k} / \sum_{k'} e^{f_{k'}}$, and one-hot labels $\mathbf{y}_i \in \{e_1, \dots, e_K\}$, where e_k is the k -th standard basis vector. Training data $S = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$ are i.i.d. The cross-entropy loss is

$$\ell_{\text{CE}}(f, \mathbf{y}) = - \sum_{k=1}^K y_k \log p_k(f).$$

A direct computation gives the well-known output-space Hessian

$$(1) \quad \nabla_f^2 \ell_{\text{CE}}(f, \mathbf{y}) = \mathbf{H}(\mathbf{x}) := \text{diag}(\mathbf{p}(\mathbf{x})) - \mathbf{p}(\mathbf{x})\mathbf{p}(\mathbf{x})^\top,$$

which is symmetric positive semidefinite, has rank $K - 1$ for any $\mathbf{p}$ in the open simplex (it has a single zero eigenvalue along $\mathbf{1}$), and rank 0 at every one-hot vertex $\mathbf{p} = e_k$ (since at $\mathbf{p} = e_k$ we have $\text{diag}(e_k) = e_k e_k^\top$, hence $\mathbf{H} = \text{diag}(e_k) - e_k e_k^\top = 0$). The Fisher matrix $\mathbf{H}(\mathbf{x})$ is the building block of the parameter Hessian:

$$(2) \quad \nabla_\theta^2 \ell_{\text{CE}}(\mathbf{x}, \mathbf{y}; \theta) = \mathbf{J}(\mathbf{x})^\top \mathbf{H}(\mathbf{x}) \mathbf{J}(\mathbf{x}) + (\text{label-dependent rank-1 term}),$$

where $\mathbf{J}(\mathbf{x}) \in \mathbb{R}^{K \times p}$ is the per-sample Jacobian of $f(\mathbf{x}; \cdot)$ with respect to θ . The label-dependent term vanishes in expectation over labels at any predictor that is exact on the population (or at any limit point with vanishing gradient), and we drop it from this point onward.

We adopt the notation of [1]. Let $\mathbf{J}_{\text{CE}} \in \mathbb{R}^{nK \times p}$ denote the stacked training Jacobian with rows $\nabla_\theta f_k(\mathbf{x}_i; \theta^*)^\top$ ranging over $(i, k) \in [n] \times [K]$, and let $\mathbf{H} = \bigoplus_{i=1}^n \mathbf{H}(\mathbf{x}_i)$ denote the block-diagonal sample Fisher. Denote the cross-class test–train cross-correlation

$$(3) \quad (\mathbf{\Gamma}_{\text{CE}})_{(j,k),(j',k')} = \mathbb{E}_{\mathbf{x}}[\psi_{j,k}(\mathbf{x})\psi_{j',k'}(\mathbf{x})], \quad \psi_{j,k}(\mathbf{x}) = \nabla_\theta f_k(\mathbf{x}; \theta^*)^\top \mathbf{v}_j,$$

where $\{\mathbf{v}_j\}$ is the right SVD basis of an appropriate factor of $\mathbf{J}_{\text{CE}}$ (see Section 3).

3. DERIVING $\mathbf{M}_{\text{CE}}$ FROM THE CROSS-ENTROPY HESSIAN

In MSE, the operator $\mathbf{M} = \mathbf{\Sigma}^{-1} \mathbf{\Gamma} \mathbf{\Sigma}^{-1}$ arises because the test risk decomposes as a quadratic form in the predicted residuals, weighted by the inverse of $\mathbf{J}^\top \mathbf{J}$ at the interpolant. Concretely, $\mathbf{\Sigma}$ comes from the SVD of the training Jacobian, and the noise-leakage variance equals $\sigma^2 \text{tr}(\mathbf{\Sigma}^{-1} \mathbf{\Gamma} \mathbf{\Sigma}^{-1})$.

For cross-entropy at a (near-)interpolating predictor, the linearized test risk is governed by the inverse of the parameter Hessian (2), that is $(\mathbf{J}_{\text{CE}}^\top \mathbf{H} \mathbf{J}_{\text{CE}})^{-1}$. Two natural factorizations arise:

- (1) Define the *Fisher-weighted Jacobian*

$$\tilde{\mathbf{J}} := \mathbf{H}^{1/2} \mathbf{J}_{\text{CE}} \in \mathbb{R}^{nK \times p},$$

with SVD $\tilde{\mathbf{J}} = \tilde{\mathbf{U}} \tilde{\mathbf{\Sigma}} \tilde{\mathbf{V}}^\top$. Then $\mathbf{J}_{\text{CE}}^\top \mathbf{H} \mathbf{J}_{\text{CE}} = \tilde{\mathbf{V}} \tilde{\mathbf{\Sigma}}^2 \tilde{\mathbf{V}}^\top$, so the parameter Hessian's eigenstructure coincides with that of $\tilde{\mathbf{J}}^\top \tilde{\mathbf{J}}$.

- (2) The right singular vectors $\tilde{\mathbf{V}}$ provide the natural basis for gradient features $\tilde{\psi}_{j,k}(\mathbf{x}) := \nabla_{\theta} f_k(\mathbf{x}; \theta^*)^\top \tilde{\mathbf{v}}_j$, and the corresponding cross-correlation matrix is $\tilde{\mathbf{\Gamma}}_{(j,k),(j',k')} = \mathbb{E}_{\mathbf{x}}[\tilde{\psi}_{j,k}(\mathbf{x}) \tilde{\psi}_{j',k'}(\mathbf{x})]$. The matrix $\tilde{\mathbf{\Sigma}}_{\text{CE}}$ refers throughout to the singular value matrix $\tilde{\mathbf{\Sigma}}$ above; these are the square roots of the eigenvalues of the parameter Hessian.

These two ingredients give the operator that propagates label noise through the linearized CE risk:

Definition 3.1 (Cross-entropy noise propagation operator). The *cross-entropy noise propagation operator* is

$$(4) \quad \mathbf{M}_{\text{CE}} := \tilde{\mathbf{\Sigma}}^{-1} \tilde{\mathbf{\Gamma}} \tilde{\mathbf{\Sigma}}^{-1}, \quad \tilde{\mathbf{J}} = \mathbf{H}^{1/2} \mathbf{J}_{\text{CE}} = \tilde{\mathbf{U}} \tilde{\mathbf{\Sigma}} \tilde{\mathbf{V}}^\top,$$

where the inverse on the right is the Moore–Penrose pseudoinverse restricted to the range of $\tilde{\mathbf{\Sigma}}$.

Definition 3.2 (Effective rank). For any non-zero positive semidefinite operator $\mathbf{A}$ on a finite-dimensional inner-product space, the *effective rank* is

$$r_{\text{eff}}(\mathbf{A}) := \frac{(\text{tr } \mathbf{A})^2}{\|\mathbf{A}\|_F^2}.$$

This quantity satisfies $1 \leq r_{\text{eff}}(\mathbf{A}) \leq \text{rank}(\mathbf{A})$, with equality on the right iff all nonzero eigenvalues are equal, and is invariant under positive scalar rescaling of $\mathbf{A}$.

Remark 3.3 (Equivalence with the form $\mathbf{F}^{-1} \mathbf{\Gamma} \mathbf{F}^{-1}$). Some prior write-ups state $\mathbf{M}_{\text{CE}} = \mathbf{F}^{-1} \mathbf{\Gamma}_{\text{CE}} \mathbf{F}^{-1}$ with $\mathbf{F} = \mathbf{H}^{1/2} \mathbf{\Sigma}_{\text{CE}}$, where $\mathbf{\Sigma}_{\text{CE}}$ denotes the singular values of $\mathbf{J}_{\text{CE}}$. This is the same operator as (4) written in the basis of $\mathbf{J}_{\text{CE}}$ rather than $\tilde{\mathbf{J}}$: $\tilde{\mathbf{\Sigma}}$ are precisely the singular values of $\mathbf{H}^{1/2} \mathbf{J}_{\text{CE}}$, which equal the square roots of the eigenvalues of $\mathbf{J}_{\text{CE}}^\top \mathbf{H} \mathbf{J}_{\text{CE}}$. We retain Definition 3.1 as the canonical form because the factorization is unambiguous.

The justification for Definition 3.1 is the following linearization lemma, which derives the noise-leakage variance from a Taylor expansion of the empirical CE risk near θ^* . The argument is the cross-entropy analog

of the MSE template in [1, §2]; the loss-specific ingredient is the substitution of the CE Hessian $\mathbf{J}^\top \mathbf{H} \mathbf{J}$ for the squared-loss Hessian $\mathbf{J}^\top \mathbf{J}$, equivalently the replacement of $\mathbf{J}$ by $\tilde{\mathbf{J}} = \mathbf{H}^{1/2} \mathbf{J}$.

Lemma 3.4 (Linearized noise leakage under CE). *Let θ^* be a stationary point of the empirical CE risk on the training set S , with parameter Hessian $\mathbf{F} := \mathbf{J}_{\text{CE}}^\top \mathbf{H} \mathbf{J}_{\text{CE}}$ invertible on its range. Suppose label perturbations $\boldsymbol{\xi}_i \in \mathbb{R}^K$ satisfy $\mathbb{E}[\boldsymbol{\xi}_i] = 0$ and $\mathbb{E}[\boldsymbol{\xi}_i \boldsymbol{\xi}_i^\top] = \sigma^2 \mathbf{H}(\mathbf{x}_i)$ (the Fisher-weighted noise model that matches the CE link). Let θ_ξ^* denote the perturbed stationary point. Then to linear order in $\boldsymbol{\xi}$,*

- (i) *the parameter perturbation is $\delta\theta = \mathbf{F}^\dagger \mathbf{J}_{\text{CE}}^\top \mathbf{H}^{1/2} \boldsymbol{\xi} = \tilde{\mathbf{V}} \tilde{\boldsymbol{\Sigma}}^{-1} \tilde{\mathbf{U}}^\top \boldsymbol{\xi}$;*
- (ii) *along any gradient feature direction $\tilde{\mathbf{v}}_j$ the variance contribution to the test residual at $\mathbf{x}_*$ equals $\sigma^2 \tilde{\boldsymbol{\Gamma}}_{jj} / \tilde{\sigma}_j^2$;*
- (iii) *summing, $\text{Var}(\text{linearized test residual}) = \sigma^2 \text{tr}(\mathbf{M}_{\text{CE}})$.*

Proof. Linearize the empirical-risk gradient near θ^* :

$$\nabla_{\theta} \hat{\mathcal{R}}_{\text{CE}}(\theta^* + \delta\theta; \mathbf{y} + \boldsymbol{\xi}) = \mathbf{F} \delta\theta - \mathbf{J}_{\text{CE}}^\top \mathbf{H}^{1/2} \boldsymbol{\xi} + O(\|\delta\theta\|^2 + \|\boldsymbol{\xi}\|^2),$$

using $\nabla_{\theta} \ell_{\text{CE}}(\mathbf{x}, \mathbf{y}; \theta) = \mathbf{J}(\mathbf{x})^\top (\mathbf{p}(\mathbf{x}; \theta) - \mathbf{y})$ and the noise model. Setting the gradient to zero gives $\delta\theta = \mathbf{F}^\dagger \mathbf{J}_{\text{CE}}^\top \mathbf{H}^{1/2} \boldsymbol{\xi}$, which is (i); substituting the SVD $\tilde{\mathbf{J}} = \tilde{\mathbf{U}} \tilde{\boldsymbol{\Sigma}} \tilde{\mathbf{V}}^\top$ into $\mathbf{F} = \tilde{\mathbf{V}} \tilde{\boldsymbol{\Sigma}}^2 \tilde{\mathbf{V}}^\top$ yields the explicit form. For (ii), the test residual contribution along $\tilde{\mathbf{v}}_j$ is $\tilde{\psi}_{j,\cdot}(\mathbf{x}_*)^\top \delta\theta$, whose variance is $\sigma^2 \tilde{\sigma}_j^{-2} \tilde{\boldsymbol{\Gamma}}_{jj}$ by orthonormality of $\tilde{\mathbf{U}}$. For (iii), sum to recognize $\sum_j \tilde{\boldsymbol{\Gamma}}_{jj} / \tilde{\sigma}_j^2 = \text{tr}(\tilde{\boldsymbol{\Sigma}}^{-1} \tilde{\boldsymbol{\Gamma}} \tilde{\boldsymbol{\Sigma}}^{-1}) = \text{tr}(\mathbf{M}_{\text{CE}})$. $\square$

This lemma plays the same role as the squared-loss derivation in [1]: $\mathbf{M}_{\text{CE}}$ is exactly the quadratic form that controls noise-leakage variance under CE, with $\mathbf{H}$ replacing the identity in the output-space inner product.

4. CROSS-ENTROPY TRICHOTOMY

Theorem 4.1 (Cross-entropy trichotomy). *Suppose the singular values $\tilde{\sigma}_j$ of $\tilde{\mathbf{J}} = \mathbf{H}^{1/2} \mathbf{J}_{\text{CE}}$ and the diagonal entries $\tilde{\boldsymbol{\Gamma}}_{jj}$ (in the right singular basis of $\tilde{\mathbf{J}}$) exhibit power-law decay $\tilde{\sigma}_j \asymp j^{-\alpha}$, $\tilde{\varrho}_j := \sqrt{\tilde{\boldsymbol{\Gamma}}_{jj}} \asymp j^{-\beta}$. Suppose furthermore that the Fisher block is uniformly non-degenerate in the sense of the operator inequality*

$$(5) \quad \mathbf{H}(\mathbf{x}_i) \succeq h_{\min} \mathbf{P}_{\mathcal{Z}_K} \quad \text{for all } i = 1, \dots, n,$$

for some $h_{\min} > 0$, where $\mathbf{P}_{\mathcal{Z}_K}$ is the orthogonal projection onto the zero-sum subspace $\mathcal{Z}_K = \{u \in \mathbb{R}^K : \mathbf{1}^\top u = 0\}$. Then

$$\text{tr}(\mathbf{M}_{\text{CE}}) < \infty \iff \beta > \alpha + \frac{1}{2}.$$

Proof. By Definition 3.1,

$$\text{tr}(\mathbf{M}_{\text{CE}}) = \sum_j \tilde{\Gamma}_{jj} / \tilde{\sigma}_j^2 \asymp \sum_j j^{-2\beta} / j^{-2\alpha} = \sum_j j^{2(\alpha-\beta)},$$

which converges iff $2(\alpha - \beta) < -1$, i.e. $\beta > \alpha + 1/2$. This is a statement about the singular values of $\tilde{\mathbf{J}}$ and the diagonal of $\tilde{\Gamma}$ in $\tilde{\mathbf{J}}$'s right singular basis, so no basis conversion is needed. Hypothesis (5) ensures that $\tilde{\mathbf{J}} = \mathbf{H}^{1/2} \mathbf{J}_{\text{CE}}$ has full rank on the zero-sum subspace whenever $\mathbf{J}_{\text{CE}}$ does, with smallest nonzero singular value at least $h_{\min}^{1/2}$ times that of $\mathbf{J}_{\text{CE}}$ on $\mathcal{Z}_K$, so the power-law hypotheses are equivalent to power laws on $\mathbf{J}_{\text{CE}}$ up to a multiplicative constant. $\square$

Remark 4.2 (On the uniform Fisher lower bound). Hypothesis (5) is the cross-entropy analog of the “well-conditioned predictor” assumption: at any predictor whose softmax outputs $\mathbf{p}(\mathbf{x})$ avoid the boundary of the simplex, the nonzero eigenvalues of $\mathbf{H}(\mathbf{x})$ are bounded below by a positive constant on $\mathcal{Z}_K$. For a stable training trajectory away from the zero-margin classifier, this is automatic with $h_{\min}$ controlled by the entropy of the outputs. The hypothesis fails in the strict NC limit; we treat that limit separately in Section 5 via a different argument.

Remark 4.3. The matrix Σ_{CE} in the form $\mathbf{M}_{\text{CE}} = \mathbf{F}^{-1} \Gamma_{\text{CE}} \mathbf{F}^{-1}$ of Remark 3.3 denotes the diagonal matrix of singular values of the (unweighted) Jacobian $\mathbf{J}_{\text{CE}}$; under hypothesis (5), the singular values of $\tilde{\mathbf{J}}$ relate to those of $\mathbf{J}_{\text{CE}}$ via $\tilde{\sigma}_j \in [h_{\min}^{1/2} \sigma_j^{\text{CE}}, h_{\max}^{1/2} \sigma_j^{\text{CE}}]$ where $h_{\max} = \|\mathbf{H}\|_{\text{op}}$.

5. EFFECTIVE RANK AT THE NEURAL COLLAPSE FIXED POINT

We now compute the effective rank $r_{\text{eff}}(\mathbf{M}_{\text{CE}})$ in the limit where the trained predictor approaches the Neural Collapse fixed point of [10]. A subtlety: in the strict infinite-margin limit (logits diverging in the simplex ETF directions), the softmax output for each sample becomes one-hot, $\mathbf{H}(\mathbf{x}) \rightarrow 0$, and $\mathbf{M}_{\text{CE}}$ degenerates. We address this by considering a sequence of predictors whose last-layer features approach the NC fixed point with *uniformly bounded logit norms*; in this regime the limiting $\mathbf{p}(\mathbf{x})$ is a probability vector strictly in the open simplex, peaked on the correct class but with positive mass on every coordinate. This is the regime relevant to NC phenomenology at any finite training time, and the one in which the unconstrained-features analyses of [7, 9, 5] yield well-defined limits. The strict-margin limit is treated separately by rescaling (Remark 5.2).

5.1. Geometry of the simplex equiangular tight frame. Recall the NC fixed point: the last-layer features satisfy $h(\mathbf{x}_i) = \mu_{c(\mathbf{x}_i)}$ where $\{\mu_c\}_{c=1}^K$ forms a simplex equiangular tight frame in $\mathbb{R}^{K-1}$ with $\sum_c \mu_c = 0$ and $\mu_c^\top \mu_{c'} = \frac{K\delta_{cc'}-1}{K-1} \|\mu\|^2$ for some common norm $\|\mu\|$. The classifier weights $\{\mathbf{w}_c\}$ likewise satisfy $\mathbf{w}_c = (\|\mathbf{w}\| / \|\mu\|)\mu_c$ up to a scalar.

The image of the per-class logit difference map $h \mapsto (\mathbf{w}_1^\top h, \dots, \mathbf{w}_K^\top h)$ at the NC fixed point therefore lies in the $(K-1)$ -dimensional zero-sum subspace

$$\mathcal{Z}_K := \{u \in \mathbb{R}^K : \mathbf{1}^\top u = 0\}.$$

Let $\mathbf{P}_Z$ be the orthogonal projection onto $\mathcal{Z}_K$, that is $\mathbf{P}_Z = \mathbf{I}_K - \frac{1}{K}\mathbf{1}\mathbf{1}^\top$.

5.2. Limiting Fisher and limiting cross-correlation.

Lemma 5.1 (Fisher matrix under bounded-logit NC). *Consider a sequence of predictors $\{\theta^{(t)}\}$ whose last-layer features approach the NC fixed point and whose logit vectors $f(\mathbf{x}_i; \theta^{(t)}) \in \mathbb{R}^K$ satisfy $\|f(\mathbf{x}_i; \theta^{(t)})\|_\infty \leq L$ for some finite $L > 0$. Then for any sample $\mathbf{x}_i$ and along any limit point of $\{\theta^{(t)}\}$, every coordinate of the predicted probability $\mathbf{p}(\mathbf{x}_i; \theta^{(t)})$ is bounded below by $p_{\min} := e^{-2L}/K > 0$, the matrix $\mathbf{H}(\mathbf{x}_i)$ has rank exactly $K-1$ with kernel $\text{span}(\mathbf{1})$, and the operator inequality*

$$\mathbf{H}(\mathbf{x}_i) \succeq h_{\min} \mathbf{P}_{\mathcal{Z}_K}, \quad h_{\min} := p_{\min}(1 - 1/K),$$

holds uniformly in i and t .

Proof. For any logit vector f with $\|f\|_\infty \leq L$, the softmax probabilities satisfy $p_k = e^{f_k} / \sum_{k'} e^{f_{k'}} \geq e^{-L} / (K e^L) = e^{-2L}/K$, giving $p_{\min} > 0$. The Fisher matrix $\mathbf{H} = \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$ satisfies $\mathbf{H}\mathbf{1} = \mathbf{p} - \mathbf{p}(\mathbf{1}^\top \mathbf{p}) = 0$, so $\text{span}(\mathbf{1}) \subseteq \ker \mathbf{H}$, with equality on the open simplex (a standard fact about the multinomial Fisher matrix; see e.g. [4, §3]). The bound on the smallest nonzero eigenvalue of $\mathbf{H}$ on $\mathcal{Z}_K$ follows from the standard estimate $\lambda_{\min}(\mathbf{H}|_{\mathcal{Z}_K}) \geq p_{\min}(1 - p_{\max}) \geq p_{\min}(1 - 1/K)$ for $\mathbf{p}$ supported on the open simplex with $p_{\max} \leq 1 - (K-1)p_{\min} \leq 1 - 1/K$ (achieved at the corner of the bounded-logit constraint set). $\square$

Remark 5.2 (Strict-one-hot limit and rescaling). In the strict infinite-margin limit ($L \rightarrow \infty$), the bounded-logit condition fails, $\mathbf{p} \rightarrow e_c$, and $\mathbf{H} \rightarrow 0$, so $\tilde{\mathbf{J}} \rightarrow 0$ and Definition 3.1 degenerates. The natural rescaled object is

$$\tilde{\mathbf{J}}_{\text{rsc}} := h_{\min}^{-1/2} \tilde{\mathbf{J}} = (h_{\min}^{-1/2} \mathbf{H}^{1/2}) \mathbf{J}_{\text{CE}},$$

which converges to the projector $\mathbf{P}_{\mathcal{Z}_K} \mathbf{J}_{\text{CE}}$ in the limit (since $h_{\min}^{-1/2} \mathbf{H}^{1/2}$ has unit spectrum on the support of $\mathbf{H}$ and zero kernel along $\mathbf{1}$). The operator $\mathbf{M}_{\text{CE}}$ rebuilt from $\tilde{\mathbf{J}}_{\text{rsc}}$ is well-defined on the zero-sum subspace and satisfies $r_{\text{eff}}(\mathbf{M}_{\text{CE}}^{\text{rsc}}) = r_{\text{eff}}(\mathbf{M}_{\text{CE}})$ since effective rank is invariant

under positive scalar rescaling of $\tilde{\Sigma}$. In what follows we work with finite logit scale where this rescaling is unnecessary.

5.3. Effective rank formula. Before stating the theorem, we make the structural NC hypotheses explicit. The simplex-ETF geometry of [10] fixes the last-layer features and classifier weights, but the per-sample Jacobian also depends on the backbone (everything below the last layer). The following assumption, standard in the unconstrained-features literature [7, 9, 5], isolates what is needed.

Assumption 5.3 (NC backbone non-degeneracy). At θ^* , with last-layer feature map $h(\mathbf{x}; \theta^*) \in \mathbb{R}^{K-1}$ and last-layer classifier $W \in \mathbb{R}^{K \times (K-1)}$ in simplex-ETF configuration:

- (a) (Class collapse / NC1) For every sample $\mathbf{x}_i$ in class $c_i \in \{1, \dots, K\}$, $h(\mathbf{x}_i; \theta^*) = \mu_{c_i}$ where $\{\mu_c\}$ is the simplex ETF ($\sum_c \mu_c = 0$, $\mu_c^\top \mu_{c'} = (K\delta_{cc'} - 1)/(K-1) \cdot \|\mu\|^2$).
- (b) (Self-duality / NC2) $W = (\|W\| / \|\mu\|) M$ where $M^\top = [\mu_1 \cdots \mu_K]$.
- (c) (Backbone full-rank in ETF directions) The backbone Jacobian $\partial h(\mathbf{x}_i; \theta^*) / \partial \theta \in \mathbb{R}^{(K-1) \times p}$ has full row rank $K-1$ for every i .

Hypothesis (c) is the chain-rule input to the per-sample Jacobian $\mathbf{J}(\mathbf{x}_i) \in \mathbb{R}^{K \times p}$; it follows in the unconstrained-features regime where h is a free variable [9], and is the operative assumption in feature-learning analyses of NC. We do not derive it from NC1–NC4; we assume it.

We measure the effective rank on the per-sample block (equivalently, on the output-averaged operator), which is the geometric object fixed by simplex-ETF symmetry and is independent of n .

Definition 5.4 (Per-sample / averaged $\mathbf{M}_{\text{CE}}$ block). Let $\mathbf{M}_{\text{CE}}^{(i)} \in \mathbb{R}^{K \times K}$ denote the per-sample restriction of $\mathbf{M}_{\text{CE}}$ to the output block indexed by sample i , viewed as an operator on $\mathbb{R}^K$. Define the *averaged operator*

$$\overline{\mathbf{M}}_{\text{CE}} := \frac{1}{n} \sum_{i=1}^n \mathbf{M}_{\text{CE}}^{(i)}.$$

Theorem 5.5 (Effective rank at NC). *Let θ^* be a limit point of the training trajectory of Lemma 5.1 satisfying Assumption 5.3, and assume the training set is class-balanced (n/K samples per class). Then both per-sample blocks and the averaged operator satisfy*

$$r_{\text{eff}}(\mathbf{M}_{\text{CE}}^{(i)}) = K - 1 \quad \text{and} \quad r_{\text{eff}}(\overline{\mathbf{M}}_{\text{CE}}) = K - 1.$$

Proof. Step 1: Per-sample block has rank exactly $K-1$ on $\mathcal{Z}_K$. By Lemma 5.1, $\mathbf{H}(\mathbf{x}_i)$ is positive definite on $\mathcal{Z}_K$ and zero on $\text{span}(\mathbf{1})$. By

Assumption 5.3(c) the backbone Jacobian has rank $K - 1$, and the chain rule gives

$$\mathbf{J}(\mathbf{x}_i) = W \cdot \frac{\partial h(\mathbf{x}_i; \theta^*)}{\partial \theta} + \left. \frac{\partial f}{\partial W} \right|_{(\mathbf{x}_i)} \cdot \frac{\partial W}{\partial \theta},$$

where $\partial f / \partial W$ at the NC weight has rank $K - 1$ on $\mathcal{Z}_K$ (the simplex-ETF rows of W span $\mathcal{Z}_K$ exactly, by Assumption 5.3(b)). Hence $\mathbf{J}(\mathbf{x}_i) \mathbf{P}_{\mathcal{Z}_K}$ has row rank exactly $K - 1$, and the same holds for $\tilde{\mathbf{J}}(\mathbf{x}_i) = \mathbf{H}(\mathbf{x}_i)^{1/2} \mathbf{J}(\mathbf{x}_i)$.

Step 2: Per-sample block has all $K - 1$ eigenvalues equal. By Lemma 5.1 and the simplex-ETF symmetry, the per-sample Fisher restricted to $\mathcal{Z}_K$ is a positive scalar multiple of the identity: $\mathbf{H}(\mathbf{x}_i)|_{\mathcal{Z}_K} = h_i \mathbf{I}_{\mathcal{Z}_K}$ for some $h_i > 0$. (Direct calculation: at NC, $\mathbf{p}(\mathbf{x}_i)$ is permutation-symmetric across the $K - 1$ non-target classes by the simplex-ETF symmetry, so $\mathbf{H}(\mathbf{x}_i)|_{\mathcal{Z}_K}$ is a scalar matrix.) Likewise, by simplex-ETF symmetry across the $K - 1$ output directions in $\mathcal{Z}_K$, the cross-correlation $\tilde{\mathbf{\Gamma}}$ restricted to a per-sample block on $\mathcal{Z}_K$ is a scalar matrix $\gamma_i \mathbf{I}_{\mathcal{Z}_K}$ for some $\gamma_i \geq 0$: the $K - 1$ within-block directions are equivalent under the action of the simplex-ETF symmetry group S_{K-1} that fixes the class of $\mathbf{x}_i$.

Step 3: Computing r_{eff} . By Steps 1–2, $\mathbf{M}_{\text{CE}}^{(i)}$ on $\mathcal{Z}_K$ is $(\gamma_i/h_i) \mathbf{I}_{\mathcal{Z}_K}$ — a positive scalar multiple of the identity on a $(K - 1)$ -dimensional space, with the remaining direction $\text{span}(\mathbf{1})$ killed by $\mathbf{H}$. All $K - 1$ nonzero eigenvalues of $\mathbf{M}_{\text{CE}}^{(i)}$ are equal, hence by Definition 3.2,

$$r_{\text{eff}}(\mathbf{M}_{\text{CE}}^{(i)}) = \frac{((K - 1) \gamma_i/h_i)^2}{(K - 1) (\gamma_i/h_i)^2} = K - 1.$$

Step 4: Averaged operator. By class-balance, the per-sample blocks group into K equal-size class subsets, each of which gives the same γ/h ratio by the simplex-ETF symmetry permuting classes. Hence $\overline{\mathbf{M}}_{\text{CE}} = (\bar{\gamma}/\bar{h}) \mathbf{I}_{\mathcal{Z}_K}$ on $\mathcal{Z}_K$ for the corresponding averages, with all $K - 1$ nonzero eigenvalues equal. By the same calculation, $r_{\text{eff}}(\overline{\mathbf{M}}_{\text{CE}}) = K - 1$. $\square$

Remark 5.6 (Dimension-free scaling for classification). Theorem 5.5 states that, at the NC fixed point, the effective rank of $\mathbf{M}_{\text{CE}}$ on each per-sample block (and of the averaged operator) depends only on K and not on the input dimension d or the parameter count p . This is the classification analog of the dimension-free regression rate (rank-one collapse onto a single feature direction): the rank- $(K - 1)$ collapse follows from the geometric fact that the simplex ETF in $\mathbb{R}^{K-1}$ spans a $(K - 1)$ -dimensional subspace, on which the Fisher matrix at NC is a scalar.

Remark 5.7 (Full-operator effective rank). The full operator $\mathbf{M}_{\text{CE}}$ on the joint sample-class space has rank $n(K - 1)$ on the zero-sum subspace by Step 1, all eigenvalues equal under class-balance and simplex-ETF symmetry, so $r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = n(K - 1)$ at the limit point. This is consistent: each of the n training samples contributes $K - 1$ equal eigenvalues. The dimension-free scaling we care about for generalization is the per-sample / averaged quantity, which captures the noise leakage *per training example*.

6. DISCUSSION

Status of the NC hypothesis. The conclusion $r_{\text{eff}}(\mathbf{M}_{\text{CE}}^{(i)}) = K - 1$ (per-sample) is conditional on the NC fixed point being attained or approached at the end of training. NC is observed empirically across diverse networks [10], and is rigorously established in the unconstrained-features model of [7, 9], which idealizes the last-layer features as free variables. The result of this paper is therefore a statement of the form “NC + backbone non-degeneracy (Assumption 5.3(c)) $\Rightarrow$ rank- $(K - 1)$ effective rank”; we do not prove that NC holds for finite networks trained end-to-end with cross-entropy. Limitations in the strict one-hot limit are addressed by rescaling (Remark 5.2); see also the discussion below.

Off-fixed-point behavior. A natural follow-up question is: how does $r_{\text{eff}}(\mathbf{M}_{\text{CE}})$ behave during the approach to the NC fixed point? We expect a gap term whose decay is governed by the convergence rate of NC, which would yield a rate of $r_{\text{eff}}(\mathbf{M}_{\text{CE}})(t) - (K - 1)$ in terms of the training time t . Quantifying this gap is open.

Class-imbalanced data. Theorem 5.5 assumes class-balanced gradient features. For imbalanced classes, the simplex ETF deforms to a class-frequency-weighted variant (see e.g. [5] for the unconstrained-features analysis), and the effective rank gains a class-frequency-dependent correction. We do not pursue this here.

Connection to the regression result of [3]. The companion paper [3] (this paper’s regression counterpart) shows that, after feature-direction alignment, the effective rank of the regression operator $\mathbf{M}$ (the squared-loss analog of $\mathbf{M}_{\text{CE}}$ in this paper) collapses to a single dimension along the target direction. In classification, the analogous collapse for $\mathbf{M}_{\text{CE}}$ is to $K - 1$ dimensions along the simplex faces: each class identifies its corresponding simplex vertex, and the rank-one regression collapse for $\mathbf{M}$ becomes a rank- $(K - 1)$ classification collapse for $\mathbf{M}_{\text{CE}}$. The two results share the same mathematical mechanism: a geometric symmetry in the trained predictor’s last-layer geometry collapses an a priori

$\Theta(d)$ rank to a small intrinsic dimension. We retain distinct notation $\mathbf{M}$ vs. $\mathbf{M}_{\text{CE}}$ throughout to disambiguate the two operators.

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REFERENCES

- [1] L. Chang. Noise propagation operators and a two-parameter characterization of benign overfitting. *Companion paper*, 2026.
- [2] L. Chang. A bias–variance decomposition for the noise propagation operator and the role of feature learning. *Companion paper*, 2026.
- [3] L. Chang. Feature learning, effective dimension, and architecture-specific critical depth. *Companion paper*, 2026.
- [4] S.-i. Amari. *Information Geometry and Its Applications*. Springer, 2016.
- [5] C. Fang, H. He, Q. Long, and W. J. Su. Exploring deep neural networks via layer-peeled model: Minority collapse in imbalanced training. *Proceedings of the National Academy of Sciences*, 118(43):e2103091118, 2021.
- [6] X. Y. Han, V. Pappas, and D. L. Donoho. Neural collapse under MSE loss: proximity to and dynamics on the central path. In *International Conference on Learning Representations (ICLR)*, 2022.
- [7] J. Lu and S. Steinerberger. Neural collapse under cross-entropy loss. *Applied and Computational Harmonic Analysis*, 59:224–241, 2022.
- [8] S. Mei, T. Misiakiewicz, and A. Montanari. Generalization error of random feature and kernel methods: Hypercontractivity and kernel matrix concentration. *Applied and Computational Harmonic Analysis*, 59:3–84, 2022.
- [9] D. G. Mixon, H. Parshall, and J. Pi. Neural collapse with unconstrained features. *Sampling Theory, Signal Processing, and Data Analysis*, 20(2):11, 2022.
- [10] V. Pappas, X. Y. Han, and D. L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020.

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